

Permutation-Symmetric Ogata Approximation Fails for Three Pauli Observables

Automated mathematics run `fa_banach_001`, lane 17

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Status. Candidate counterexample, likely valid, requiring human review. The result is a full negative answer to the simultaneous tensor-permutation-symmetry instantiation of the source question; it does not settle every other symmetry pattern mentioned there.

Abstract

Herrera asks whether the commuting approximants in Ogata’s theorem can preserve the symmetries of three or more macroscopic observables. We show that the most canonical tensor-permutation version is false. For the three macroscopic Pauli observables on N qubits, every commuting self-adjoint triple invariant under the transposition $(1\ 2)$ and the long cycle $(1\ 2\ \dots\ N)$ has maximum coordinate error at least $1/6$, independently of N . The two symmetries generate S_N , so the approximants reduce the fully symmetric tensor block. There the observables form the normalized irreducible spin- $N/2$ triple. Its Pauli localizer has unbalanced inertia, whereas the invertible localizer of a commuting triple has balanced inertia. The argument is elementary and gives the explicit constant.

1 Source question and exact scope

The source is David Herrera, *A Projection Characterization and Symmetry Bootstrap for Elements of a von Neumann Algebra that are Nearby Commuting Elements*, arXiv:2412.20795v1 (2024) [1]. In Example 2.5, PDF pages 14–15, it defines the tensor-factor permutation symmetries $S_\sigma(A) = U_\sigma A U_\sigma^{-1}$ on $M_d(\mathbb{C})^{\otimes N}$ and notes that the macroscopic observable

$$T_N(A) = \frac{1}{N} \sum_{r=1}^N I_d^{\otimes(r-1)} \otimes A \otimes I_d^{\otimes(N-r)}$$

is fixed by every S_σ . It also notes that fixedness for the transposition $(1\ 2)$ and long cycle $(1\ 2\ \dots\ N)$ already implies fixedness for all permutations.

The question is transcribed as follows:

“A natural question that one can ask is if a symmetry version of Ogata’s theorem holds. Namely, when H_i satisfy certain symmetries and antisymmetries, is it true that the H_i are nearby commuting matrices that also satisfy these symmetries. How much symmetry that is preserved for $m \geq 3$ is still not known.”

Our scope is precise: can the two noncommuting permutation generators be preserved *simultaneously* for the three macroscopic Pauli observables? The answer is no. We do not claim that preserving one fixed S_σ at a time is impossible.

A natural question that one can ask is if a symmetry version of Ogata’s theorem holds. Namely, when H_i satisfy certain symmetries and antisymmetries, is it true that the H_i are nearby commuting matrices that also satisfy these symmetries.

How much symmetry that is preserved for $m \geq 3$ is still not known. One can readily show from Lin’s theorem with linear symmetries (Theorem 4.5) that the nearby commuting matrices can be chosen to be S_σ -symmetric when $m = 2$.

In [17], we showed that when $m = 3$ and $d = 2$ then complex conjugation-symmetry can be preserved. By Remark 1.12, our Symmetry Bootstrap also shows that conjugation symmetry can also be preserved when $m = 3$ and $d \geq 2$ as a direct consequence of Ogata’s original proof if one of the matrices H_i is purely imaginary and the others

Figure 1: The open question in arXiv:2412.20795v1, PDF page 15.

2 Definitions and proof intuition

Let

$$s_1 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad s_2 = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad s_3 = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

be the norm-1/2 Pauli spin matrices, and set $H_{i,N} = T_N(s_i)$ on $\mathcal{H}_N = (\mathbb{C}^2)^{\otimes N}$. Write U_σ for the unitary permuting tensor factors according to $\sigma \in S_N$.

For self-adjoint X_1, X_2, X_3 on a finite-dimensional Hilbert space, define their Pauli localizer

$$\mathcal{L}(X) = \sum_{i=1}^3 X_i \otimes \tau_i, \quad \tau_i = 2s_i.$$

The τ_i are the unit-norm Pauli matrices.

Proof intuition. Ogata’s theorem permits its commuting approximants to mix the spin sectors in the tensor power. Full permutation invariance removes that escape route. Indeed, any permutation-invariant operator reduces the symmetric tensor subspace $K_N = \text{Sym}^N(\mathbb{C}^2)$. The compressed macroscopic Pauli triple on K_N is the “fuzzy sphere,” or normalized spin- $N/2$ triple. Its localizer has one extra positive channel: positive and negative multiplicities are $N + 2$ and N . By contrast, a commuting triple has joint eigenvalues $z \in \mathbb{R}^3$, and each nonzero z contributes the balanced pair $\{+|z|, -|z|\}$ to its localizer. Since the fuzzy-sphere localizer has a uniform gap 1/2 about zero, this inertia mismatch cannot disappear under a small perturbation.

3 Counterexample theorem

Theorem 1 (uniform obstruction). *For every integer $N \geq 2$, let $H_{i,N} = T_N(s_i)$, $i = 1, 2, 3$. Suppose $Y_{1,N}, Y_{2,N}, Y_{3,N} \in M_{2N}(\mathbb{C})$ are self-adjoint and pairwise commuting, and suppose each $Y_{i,N}$ is fixed under conjugation by both $U_{(12)}$ and $U_{(12\dots N)}$. Then*

$$\max_{1 \leq i \leq 3} \|Y_{i,N} - H_{i,N}\| \geq \frac{1}{6}.$$

Consequently, the macroscopic Pauli triple admits no asymptotically close commuting self-adjoint approximants preserving those two symmetries simultaneously.

The proof uses two elementary lemmas.

Lemma 1 (symmetric-block localizer). *Let $K_N = \text{Sym}^N(\mathbb{C}^2)$ and let $X_i = H_{i,N}|_{K_N}$. Then $\dim K_N = N + 1$ and the spectrum of $\mathcal{L}(X)$ is*

$$\left\{ \frac{1}{2} \text{ with multiplicity } N + 2, \quad -\left(\frac{1}{2} + \frac{1}{N}\right) \text{ with multiplicity } N \right\}.$$

In particular, $\mathcal{L}(X)$ is invertible, its distance from singular self-adjoint matrices is $1/2$, and its inertia is $(N + 2, N)$.

Proof. The symmetric power K_N carries the irreducible spin $j = N/2$ representation. If J_1, J_2, J_3 are its generators, then $X_i = J_i/N$. We compute the localizer directly, avoiding any index theorem.

Choose the J_3 -basis $\{|m\rangle : -j \leq m \leq j\}$, so

$$J_3|m\rangle = m|m\rangle, \quad J_-|m\rangle = \sqrt{(j+m)(j-m+1)}|m-1\rangle.$$

Writing $\tau_{\pm} = (\tau_1 \pm i\tau_2)/2$, we have

$$N\mathcal{L}(X) = J_3 \otimes \tau_3 + J_+ \otimes \tau_- + J_- \otimes \tau_+.$$

The vectors $|j\rangle \otimes |\uparrow\rangle$ and $|-j\rangle \otimes |\downarrow\rangle$ are eigenvectors with eigenvalue j . For each $m = -j + 1, \dots, j$, the span of $|m\rangle \otimes |\downarrow\rangle$ and $|m-1\rangle \otimes |\uparrow\rangle$ is invariant, and the matrix of $N\mathcal{L}(X)$ there is

$$\begin{pmatrix} -m & c_m \\ c_m & m-1 \end{pmatrix}, \quad c_m^2 = (j+m)(j-m+1) = j(j+1) - m(m-1).$$

Its trace is -1 and its determinant is $-j(j+1)$, so its eigenvalues are j and $-(j+1)$. There are $2j = N$ such two-dimensional blocks, plus the two extreme eigenvectors. Dividing by $N = 2j$ gives the stated spectrum and multiplicities. $\square$

Lemma 2 (commuting triples have balanced inertia). *If Z_1, Z_2, Z_3 are commuting self-adjoint operators on an n -dimensional complex Hilbert space and $\mathcal{L}(Z)$ is invertible, then its inertia is (n, n) .*

Proof. Simultaneously diagonalize the Z_i . For every joint eigenvalue $z = (z_1, z_2, z_3) \in \mathbb{R}^3$, the corresponding two-dimensional block of $\mathcal{L}(Z)$ is $z_1\tau_1 + z_2\tau_2 + z_3\tau_3$, whose eigenvalues are $\pm|z|$. Invertibility excludes $z = 0$, so every joint eigenspace contributes equally to the positive and negative counts. $\square$

Proof of Theorem 1. The transposition and long cycle generate S_N . Fixedness under conjugation by their unitaries therefore implies $[Y_{i,N}, U_\sigma] = 0$ for every $\sigma \in S_N$. The group average

$$P_N = \frac{1}{N!} \sum_{\sigma \in S_N} U_\sigma$$

is the orthogonal projection onto $K_N = \text{Sym}^N(\mathbb{C}^2)$, hence every $Y_{i,N}$ commutes with P_N . The restrictions $Z_i = Y_{i,N}|_{K_N}$ are consequently self-adjoint and pairwise commuting, and

$$\|Z_i - X_i\| \leq \|Y_{i,N} - H_{i,N}\|.$$

Assume for contradiction that the maximum on the right is less than $1/6$. Since $\|\tau_i\| = 1$,

$$\|\mathcal{L}(Z) - \mathcal{L}(X)\| \leq \sum_{i=1}^3 \|Z_i - X_i\| < \frac{1}{2}.$$

Lemma 1 says that $\mathcal{L}(X)$ has spectral gap $1/2$ about zero. Thus the straight-line path from $\mathcal{L}(X)$ to $\mathcal{L}(Z)$ remains invertible, so inertia is constant along it. Hence $\mathcal{L}(Z)$ would have inertia $(N + 2, N)$. But the Z_i commute on the $(N + 1)$ -dimensional space K_N , so Lemma 2 gives inertia $(N + 1, N + 1)$, a contradiction. $\square$

4 Checks, limitations, and novelty status

Independent calculation. The included script `code/verify_localizer.py` constructs the spin matrices and checks the exact spectrum, Casimir relation, commutators, and gap numerically for $1 \leq N \leq 24$. Errors were below $3 \cdot 10^{-16}$. This is only a consistency check; the proof above is exact.

What is and is not disproved. The theorem answers negatively the simultaneous preservation of the two specific linear multiplicative symmetries which the source identifies as generators of all tensor permutations. It therefore also rules out simultaneously preserving all S_N symmetries. It does not rule out preservation of one fixed permutation symmetry at a time. Nor does it decide the source’s transpose, conjugation, real, symplectic, phase, or signed symmetry patterns. Ordinary Ogata approximation remains valid: the counterexample shows only that its approximants must break enough tensor-permutation symmetry to couple the obstructed sector to other sectors.

Bounded novelty audit. The run indexes contained no duplicate. On 11 August 2026, OpenAlex reported zero citing works for arXiv:2412.20795. Exact-phrase searches for the source question and close searches combining Ogata approximation, tensor-permutation invariance, symmetric powers, spin matrices, Bott index, and localizers found no direct answer. We inspected the source, Ogata’s theorem [2], Herrera’s constructive qubit treatment [3], and standard Bott/localizer sources [4, 5]. Those sources supply context for the obstruction but do not state this application to simultaneous permutation symmetry. The novelty assessment is therefore moderate-to-high under a bounded search, not an exhaustive priority claim.

Human-review recommendation. Prioritize review. Verify (i) that simultaneous preservation of the two generators is within the intended scope of Example 2.5, (ii) the reduction to $\text{Sym}^N(\mathbb{C}^2)$, and (iii) the two localizer inertia calculations. No conditional lemma remains.

References

- [1] D. Herrera, *A Projection Characterization and Symmetry Bootstrap for Elements of a von Neumann Algebra that are Nearby Commuting Elements*, arXiv:2412.20795v1 (2024), Example 2.5.
- [2] Y. Ogata, *Approximating macroscopic observables in quantum spin systems with commuting matrices*, J. Funct. Anal. **264** (2013), 2005–2033; arXiv:1111.5933.
- [3] D. Herrera, *Constructing Nearby Commuting Matrices for Reducible Representations of $su(2)$ with an Application to Ogata’s Theorem*, arXiv:2212.06012v2 (2023).
- [4] M. B. Hastings and T. A. Loring, *Almost Commuting Matrices, Localized Wannier Functions, and the Quantum Hall Effect*, J. Math. Phys. **51** (2010), 015214; arXiv:0910.5490.
- [5] T. A. Loring, *A Guide to the Bott Index and Localizer Index*, arXiv:1907.11791 (2019).